

Unit 36: Using Basic Integration Formulas — Full Solutions

Warm-up

$$1. \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$$

$$2. \int \frac{dx}{1+x^2} = \arctan x + C$$

$$3. \int x^4 dx = \frac{x^5}{5} + C$$

$$4. \int e^{5x} dx = \frac{1}{5}e^{5x} + C$$

$$5. \int \cos(4x) dx = \frac{1}{4}\sin(4x) + C$$

$$6. \int \frac{dx}{\sqrt{4-x^2}}$$

Using $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$ with $a = 2$:

$$= \arcsin\left(\frac{x}{2}\right) + C$$

Standard

$$7. \int \frac{dx}{9+x^2}$$

Using $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$ with $a = 3$:

$$= \frac{1}{3}\arctan\left(\frac{x}{3}\right) + C$$

$$8. \int \frac{dx}{\sqrt{16-x^2}}$$

With $a = 4$:

$$= \arcsin\left(\frac{x}{4}\right) + C$$

$$9. \int \frac{3dx}{x^2+4}$$

With $a = 2$:

$$= 3 \cdot \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C = \frac{3}{2}\arctan\left(\frac{x}{2}\right) + C$$

10. $\int \frac{dx}{\sqrt{9-4x^2}}$

Let $u = 2x$, $du = 2 dx \Rightarrow dx = \frac{du}{2}$. Rewrite $9 - 4x^2 = 9 - u^2$.

$$\int \frac{1}{\sqrt{9-u^2}} \cdot \frac{du}{2} = \frac{1}{2} \arcsin\left(\frac{u}{3}\right) + C = \frac{1}{2} \arcsin\left(\frac{2x}{3}\right) + C$$

11. Complete the square: $x^2 - 6x + 13$.

Take half of -6 (giving -3), square it (giving 9):

$$x^2 - 6x + 13 = (x^2 - 6x + 9) - 9 + 13 = (x-3)^2 + 4$$

12. $\int \frac{dx}{x^2 - 6x + 13}$

Using the result from Problem 11: $x^2 - 6x + 13 = (x-3)^2 + 4$. Let $u = x-3$, $du = dx$.

$$\int \frac{du}{u^2 + 4} = \frac{1}{2} \arctan\left(\frac{u}{2}\right) + C = \frac{1}{2} \arctan\left(\frac{x-3}{2}\right) + C$$

13. $\int \frac{dx}{\sqrt{-x^2 + 2x + 3}}$

Complete the square: $-x^2 + 2x + 3 = -(x^2 - 2x) + 3 = -(x^2 - 2x + 1 - 1) + 3 = -(x-1)^2 + 1 + 3 = 4 - (x-1)^2$.

Let $u = x-1$, $du = dx$.

$$\int \frac{du}{\sqrt{4-u^2}} = \arcsin\left(\frac{u}{2}\right) + C = \arcsin\left(\frac{x-1}{2}\right) + C$$

Challenge

14. $\int \frac{dx}{\sqrt{-x^2 + 4x - 3}}$

Complete the square: $-x^2 + 4x - 3 = -(x^2 - 4x) - 3 = -(x^2 - 4x + 4 - 4) - 3 = -(x-2)^2 + 4 - 3 = 1 - (x-2)^2$.

Let $u = x-2$, $du = dx$.

$$\int \frac{du}{\sqrt{1-u^2}} = \arcsin(u) + C = \arcsin(x-2) + C$$

15. $\int \frac{dx}{x^2 + 4x + 13}$

Complete the square: $x^2 + 4x + 13 = (x^2 + 4x + 4) - 4 + 13 = (x+2)^2 + 9$.

Let $u = x + 2$, $du = dx$.

$$\int \frac{du}{u^2 + 9} = \frac{1}{3} \arctan\left(\frac{u}{3}\right) + C = \frac{1}{3} \arctan\left(\frac{x+2}{3}\right) + C$$

16. $\int \frac{dx}{\sqrt{8-2x-x^2}}$

Complete the square: $8 - 2x - x^2 = -(x^2 + 2x) + 8 = -(x^2 + 2x + 1 - 1) + 8 = -(x+1)^2 + 1 + 8 = 9 - (x+1)^2$.

Let $u = x + 1$, $du = dx$.

$$\int \frac{du}{\sqrt{9-u^2}} = \arcsin\left(\frac{u}{3}\right) + C = \arcsin\left(\frac{x+1}{3}\right) + C$$

17. $\int \frac{(2x+3)dx}{x^2+4}$

Split into two separate integrals:

$$\int \frac{2x}{x^2+4} dx + \int \frac{3}{x^2+4} dx$$

First piece: let $w = x^2 + 4$, $dw = 2x dx$.

$$\int \frac{dw}{w} = \ln|w| + C = \ln(x^2 + 4) + C \quad (\text{no absolute value needed, since } x^2 + 4 > 0 \text{ always})$$

Second piece: using $a = 2$:

$$3 \cdot \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C = \frac{3}{2} \arctan\left(\frac{x}{2}\right) + C$$

Combine:

$$\int \frac{(2x+3)dx}{x^2+4} = \ln(x^2+4) + \frac{3}{2} \arctan\left(\frac{x}{2}\right) + C$$

18. $\int \frac{dx}{\sqrt{3-2x-x^2}}$

Complete the square: $3 - 2x - x^2 = -(x^2 + 2x) + 3 = -(x^2 + 2x + 1 - 1) + 3 = -(x+1)^2 + 1 + 3 = 4 - (x+1)^2$.

Let $u = x + 1$, $du = dx$.

$$\int \frac{du}{\sqrt{4-u^2}} = \arcsin\left(\frac{u}{2}\right) + C = \arcsin\left(\frac{x+1}{2}\right) + C$$

Applied

19. $v(t) = \frac{1}{\sqrt{4-t^2}}$, $s(0) = 0$.

$$s(t) = \int \frac{dt}{\sqrt{4-t^2}} = \arcsin\left(\frac{t}{2}\right) + C$$

Apply $s(0) = 0$: $\arcsin(0) + C = 0 \Rightarrow C = 0$.

$$s(t) = \arcsin\left(\frac{t}{2}\right).$$

20. $I(x) = \frac{1}{x^2+9}$, find $\int_0^3 I(x) dx$.

$$\int_0^3 \frac{dx}{x^2+9} = \left[\frac{1}{3} \arctan\left(\frac{x}{3}\right) \right]_0^3 = \frac{1}{3} \arctan(1) - \frac{1}{3} \arctan(0)$$

$$= \frac{1}{3} \cdot \frac{\pi}{4} - 0 = \frac{\pi}{12}$$

Answer: $\frac{\pi}{12}$.

21. $\int \frac{dx}{\sqrt{-x^2+6x-5}}$

Complete the square: $-x^2+6x-5 = -(x^2-6x)-5 = -(x^2-6x+9-9)-5 = -(x-3)^2+9-5 = 4-(x-3)^2$.

Let $u = x-3$, $du = dx$.

$$\int \frac{du}{\sqrt{4-u^2}} = \arcsin\left(\frac{u}{2}\right) + C = \arcsin\left(\frac{x-3}{2}\right) + C$$

22. $\int_{-2}^2 \frac{dx}{\sqrt{4-x^2}}$

$$\left[\arcsin\left(\frac{x}{2}\right) \right]_{-2}^2 = \arcsin(1) - \arcsin(-1) = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

Answer: π .